

Marta Agnieszka Mrozowska

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Work Experience

Postdoctoral researcher August 2025 – present

Atmospheric and Oceanic Sciences, Princeton University

Project: Multiscale Machine Learning In Coupled Earth System Modeling ([M2LInES](#))

Lead PI: Laure Zanna

Postdoctoral researcher January 2025 – July 2025

Niels Bohr Institute, University of Copenhagen

Project: Climate Tipping Points ([ClimTip](#))

Lead PIs: Niklas Boers, Sebastian Bathiany

PhD in Physics 2021 – 2025

Niels Bohr Institute, University of Copenhagen

Project: Next Generation of Earth Modeling Systems ([NextGEMS](#))

Lead PIs: Bjorn Stevens, Irina Sandu

Thesis: *Boundary layer turbulence: Exploring strategies to improve mixing parameterizations in ocean models* [\[PDF\]](#)

Student Instructor at Ungdomslaboratoriet (ULAB) 2016 – 2021

Niels Bohr Institute, University of Copenhagen

As a member of the ULAB team, I instructed pupils from 9th-10th grade, high school and exchange university students in lab experiments offered by the Institute. Together with the rest of the team, we organized and planned laboratory exercises to inspire our guests to get interested in and explore physics.

Education

MSc in Physics 2019 – 2021

Niels Bohr Institute, University of Copenhagen

Thesis: *Hitting rock bottom: How the Earth system resists cooling and/or lowering atmospheric CO₂ concentrations during the glacial periods* [\[PDF\]](#)

BSc in Physics 2015 – 2019

Niels Bohr Institute, University of Copenhagen

Thesis: *On the connection between upwelling, opal deposit rates and atmospheric CO₂ concentration* [\[PDF\]](#)

Publications

- [1] **Mrozowska, M. A.**, Avery, J., Stoustrup, A., Nuterman, R., Johnsen, C.-J., Thormann, A., & Jochum, M. (2025). Bayesian optimization with GPU acceleration for ocean models. *Journal of Geophysical Research: Machine Learning and Computation*, 2(3), e2024JH000517. [doi:10.1029/2024JH000517](https://doi.org/10.1029/2024JH000517)
- [2] Bastin, S., Koldunov, A., Schütte, F., Gutjahr, O., **Mrozowska, M. A.**, Fischer, T., et al. (2025). Sensitivity of the tropical Atlantic to vertical mixing in two ocean models (ICON-O v2.6.6 and FESOM v2.5). *Geoscientific Model Development*, 18(4), 1189–1220. [doi:10.5194/gmd-18-1189-2025](https://doi.org/10.5194/gmd-18-1189-2025)
- [3] **Mrozowska, M. A.**, Jochum, M., Bastin, S., Hummels, R., Koldunov, A., Dengler, M., et al. (2024). Using NIW observations to assess mixed layer parameterizations: A case study in the tropical Atlantic. *Journal of Geophysical Research: Oceans*, 129, e2024JC020985. [doi:10.1029/2024JC020985](https://doi.org/10.1029/2024JC020985)

Talks

- [1] **Mrozowska, M. A.**, Reichl, B., & Adcroft, A. (2026, June). Hybrid vertical mixing parameterizations for ocean models. *Community Earth System Model 2026 Annual Workshop*.
- [2] **[Invited keynote talk] Mrozowska, M. A.**, Stoustrup, I., Avery, J., Johnsen, C.-J., Nuterman, R., & Jochum, M. (2025, January). Bayesian optimization of ocean models. *DRAKKAR Workshop 2025*. [\[Abstract\]](#) [\[Video\]](#)
- [3] **Mrozowska, M. A.**, Stoustrup, I., Avery, J., Johnsen, C.-J., Nuterman, R., & Jochum, M. (2024, September). Bayesian optimisation of ocean models. *Oxford Workshop on Model Uncertainty*. [\[Abstract\]](#)
- [4] **[Invited talk] Mrozowska, M. A.**, Stoustrup, I., Avery, J., Johnsen, C.-J., Nuterman, R., & Jochum, M. (2024, September). Bayesian optimization of ocean mixed layer parameterizations. *Physical Oceanography Group at Oxford University*.
- [5] **[Invited talk] Mrozowska, M. A.** (2024, July). In search of the optimal vertical mixing scheme for the tropical ocean. *Leibniz Institute for Baltic Sea Research Warnemünde*.
- [6] **[Invited talk] Mrozowska, M. A.**, Stoustrup, I., Avery, J., Johnsen, C.-J., Nuterman, R., & Jochum, M. (2024, April). Bayesian optimization of ocean mixed layer parameterizations. *LEAP ML Journal Club*.
- [7] **Mrozowska, M. A.**, Jochum, M., Ranum Hansen, R., & Nuterman, R. (2022, July). Near-inertial wave amplitude in the tropics as a diagnostic for validating mixed layer dynamics in general circulation models. *3rd Summer School on Theory, Mechanisms and Hierarchical Modeling of Climate Dynamics: Tropical Oceans, ENSO and their Teleconnections*, ICTP. [\[Abstract\]](#)

Posters

- [1] **Mrozowska, M. A.**, Reichl, B., & Adcroft, A. (2026, February). A data-driven parameterization of potential energy conversion rate of vertical mixing due to Langmuir turbulence. *Ocean Sciences Meeting 2026*, AGU. [\[Abstract\]](#)

- [2] Mouttapa, G., Le Sommer, J., **Mrozowska, M. A.**, Cosme, E., Wagner, G., & Deremble, B. (2026, February). Comparison of three data-driven methods for the calibration of sub-grid parametrizations. *Ocean Sciences Meeting 2026*, AGU. [\[Abstract\]](#)
- [3] **Mrozowska, M. A.**, Stoustrup, A., Avery, J., Johnsen, C.-J., Nuterman, R., & Jochum, M. (2025, June). Bayesian optimization of GPU-accelerated ocean models. *Gordon Research Conference: Machine Learning for Actionable Climate Science 2025*.
- [4] **Mrozowska, M. A.**, Jochum, M., Avery, J., Stoustrup, I., & Nuterman, R. (2024, April). Bayesian optimization of ocean mixed layer parameterizations. *EGU General Assembly 2024*. [doi:10.5194/egusphere-egu24-15235](https://doi.org/10.5194/egusphere-egu24-15235)
- [5] **Mrozowska, M. A.**, Jochum, M., Ranum Hansen, R., & Nuterman, R. (2022, July). Near-inertial wave amplitude in the tropics as a diagnostic for validating mixed layer dynamics in general circulation models. *3rd Summer School on Theory, Mechanisms and Hierarchical Modeling of Climate Dynamics: Tropical Oceans, ENSO and their Teleconnections*, ICTP.

Mentorship

- [1] Thormann, A. (2025, January). Optimizing eddy kinetic energy in Veros. *Bachelor Thesis supervised by Markus Jochum and **Marta A. Mrozowska***, University of Copenhagen. [\[PDF\]](#)

Cruise Reports and Datasets

- [1] **Mrozowska, M. A.** (2025). Bayesian optimization with GPU acceleration for ocean models. *University of Copenhagen*. [doi:10.17894/UCPH.F1DF2C14-D466-4A16-AB8A-252614FBFDFB](https://doi.org/10.17894/UCPH.F1DF2C14-D466-4A16-AB8A-252614FBFDFB)
- [2] Dengler, M., Arévalo-Martínez, D. L., Aroucha, L. C., Awo, F. M., Begler, C., Eisnecker, P., et al. (2024). Benguela Niños: Physical processes in upwelling regions and long-term variability, Cruise No. M189/1, April 16 – May 13, 2023, Walvis Bay, Namibia (Vol. M189/1). Bonn: Begutachtungspanel Forschungsschiffe. [doi:10.48433/cr_m189_1](https://doi.org/10.48433/cr_m189_1)

Software

- [1] Stoustrup, A. & **Mrozowska, M. A.** (2026). Veropt v1.2.2. *Zenodo*. [doi:10.5281/zenodo.19119716](https://doi.org/10.5281/zenodo.19119716)

Teaching

- Guest lecture on geophysical turbulence in Fluid Dynamics (BSc level) Spring 2025
Niels Bohr Institute, University of Copenhagen
- Guest lecture on Bayesian optimization Summer 2024
Bornö Summer Workshop
- Guest lecture on Ocean physics and turbulence closures Summer 2023
Bornö Summer Workshop
- TA in Fluid Dynamics (BSc level) Spring 2023
Niels Bohr Institute, University of Copenhagen

- TA in Fluid Dynamics (BSc level) Spring 2022
Niels Bohr Institute, University of Copenhagen
- TA in Computational Atmosphere and Ocean Dynamics (MSc level) Spring 2022
Niels Bohr Institute, University of Copenhagen

Outreach

- Aarhus Folkeuniversitetet: Spring 2024
Mysteries of the Deep Ocean
- Latest Thinking [\[Video\]](#): 2024
How Well Do High Resolution Ocean Models Simulate Local Turbulent Mixing?
- Science Explainer [\[PDF\]](#): 2023
Tackling the biases of tropical mixed layer depth
- Copenhagen Culture Night: Fluid Dynamics October 2022
- *SEO Summer Slackathon* [\[Article\]](#) August 2022
- Copenhagen Culture Night: Fish Don't Like It Hot October 2021
- Copenhagen Culture Night: ULAB October 2019
- Copenhagen Culture Night: ULAB October 2018

Service

- Science Board Member: DRAKKAR Workshop 2026 – present
- Organizer: M2LInES Early Career Researcher meetings 2026 – present
- Discussion Co-Converner: Climate model calibration, DRAKKAR Workshop 2025
- Organizer: University of Copenhagen PhD Journal Club 2023 – 2025

Reviewer

- Reviewer (articles reviewed): EGU Hydrology and Earth System Sciences (1), EGU Earth System Dynamics (1), Journal of Geophysical Research: Oceans (1)

Personal Development

- [Princeton Open Hackathon](#): Team Hurritrain Spring 2026
- Atmospheric and Oceanic Sciences, Princeton University:
Special topics in ocean tracer methods Fall 2025
- Niels Bohr Institute, University of Copenhagen:
[Between models and reality: School on Machine Learning in Physics](#) Summer 2025

- PhD courses, University of Copenhagen:

Machine Learning	Fall 2023
Scientific Computing	Fall 2022
- Bornö Summer Workshops:

The nexus of resolution, biogeochemistry and machine learning	Summer 2024
NextGEMS – alternative approaches	Summer 2023
The tropical mixed layer – real and simulated	Summer 2022
Carbon Cycle Workshop	Summer 2021
Turbulence Workshop	Summer 2021
Ocean Waves	Summer 2019